



Barker College

Barker Student Number: _____

Mathematics Extension 1

Staff Involved:

- GDH*
- WMD*
- JM
- BTP
- GIC
- LJP
- CFR

65 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages
- Board-approved calculators may be used
- A table of standard integrals is provided on page 9
- ALL necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged working

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

2006 TRIAL HIGHER SCHOOL CERTIFICATE

PM FRIDAY 11 AUGUST

Question 1 (12 marks) [BEGIN A NEW PAGE]

Marks

- (a) Point A has coordinates $(-2, 4)$. Point B has coordinates $(10, -8)$. Find the coordinates of the point P that divides the interval AB **externally** in the ratio $3 : 2$.

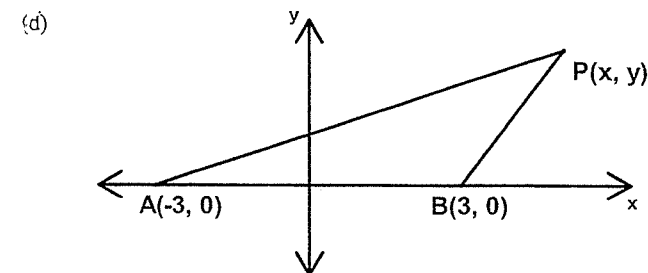
2

- (b) Find $\int x\sqrt{2x-1} dx$ using the substitution $u = 2x - 1$.

3

- (c) State whether the following claim is true or false and give a reason why:
“Because there are two types of students at Barker (day and boarder), the probability that a randomly selected Barker student is a boarder is 50%.”

1



The locus of P follows this rule:

“The gradient of PA is one unit less than the gradient of PB.”

- (i) Show that this locus has equation: $x^2 = 6y + 9$.

2

- (ii) Hence, or otherwise, find the coordinates of the focus of this locus.

1

- (e) Solve: $\frac{1}{x^3} > \frac{1}{x^5}$.

3

Question 2 (12 marks) [BEGIN A NEW PAGE]

- (a) By writing $\sin(-15^\circ)$ in the form $\sin(A - B)$, find the exact value of $\sin(-15^\circ)$.

Marks

2

- (b) Consider the functions $f(x) = \cos^{-1}(2x)$ and $g(x) = \sin^{-1}x$.

- (i) Sketch $f(x) = \cos^{-1}(2x)$.

2

- (ii) Prove that the x -coordinate of the point of intersection of $f(x)$ and $g(x)$ is $\frac{1}{\sqrt{5}}$.

2

- (iii) Show that the gradients of the tangents to $f(x)$ and $g(x)$ at their point of intersection are $-2\sqrt{5}$ and $\frac{\sqrt{5}}{2}$ respectively.

3

- (iv) Write the expansion of $\tan(\beta - \alpha)$.

1

- (v) Hence, or otherwise, find the acute angle between $f(x)$ and $g(x)$ at their point of intersection (to the nearest degree).

2

Question 3 (11 marks) [BEGIN A NEW PAGE]

Marks

- (a) (i) If $y = \ln(\sin x)$, find $\frac{dy}{dx}$.

1

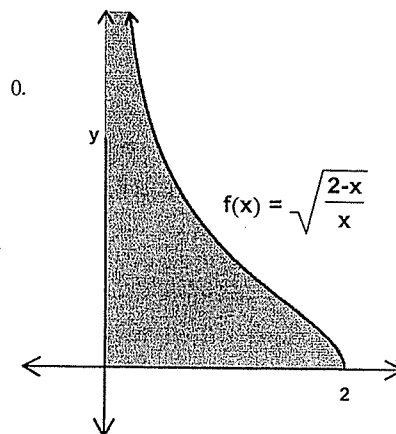
- (ii) Hence, or otherwise, find $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \cot x \, dx$.

2

- (b) Find $\int_0^{\frac{\pi}{12}} \cos^2(3x) \, dx$.

3

- (c) The curve on the right has an asymptote at $x = 0$. The shaded area 'goes forever' but its value has a limit. Steffi Graph was trying to find this limit but was unable to integrate $f(x)$. Her friend Monica suggested she use the inverse function to find the shaded area.



- (i) State the domain and range of the inverse function $f^{-1}(x)$.

1

- (ii) Roughly sketch the inverse function $f^{-1}(x)$.

1

- (iii) Show that $f^{-1}(x) = \frac{2}{1+x^2}$.

2

- (iv) Hence, or otherwise, find the limit that the value of the shaded area approaches.

2

Question 4 (12 marks) [BEGIN A NEW PAGE]

Marks

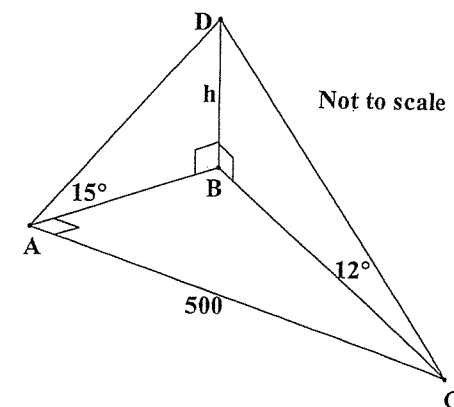
- (a) Consider the curve: $y = \frac{x^2 - 3}{x + 2}$
- (i) Find all intercepts and the equation of the vertical asymptote. 2
- (ii) Find and determine the nature of the stationary points. 3
- (iii) Show that $(x - 2) + \frac{1}{x + 2} = \frac{x^2 - 3}{x + 2}$. 1
- (iv) Hence, or otherwise, find the equation of any non-vertical asymptotes by considering what happens as $x \rightarrow \pm\infty$. 1
- (v) Sketch the curve showing all the above features (you can assume there are no points of inflexion). 1
- (b) Prove, by mathematical induction, that $5^n + 3$ is divisible by 2 for $n \geq 0$ where n is an integer. 4

Question 5 (12 marks) [BEGIN A NEW PAGE]

Marks

- (a) A particle moves in a straight line and its position in metres at any time t seconds is given by the equation: $x = 5\cos(2t) - 12\sin(2t)$.
- (i) Show, by differentiation, that the motion is simple harmonic. 2
- (ii) State the period of the motion. 1
- (iii) Express x in the form $R\cos(2t + \alpha)$. 2
- (iv) Hence, or otherwise, state the amplitude of the motion. 1
- (v) Find the general solution for all the times when the particle is at the centre of the motion. 2

- (b) From a point A, the top of a tower BD directly north of A has an angle of elevation of 15° . After walking 500 metres on a bearing of 90° , the top of the tower has an angle of elevation of 12° . Let h be the height of the tower.



- (i) Give an expression for AB in terms of h . 1
- (ii) Hence, find the height of the tower (to the nearest metre). 3

Question 6 (12 marks) [BEGIN A NEW PAGE]

Marks

- (a) Newton's Law of Cooling can be written with t as the subject as follows:

$$t = -\frac{1}{A} \ln \left(\frac{B - C}{D} \right).$$

Showing steps of working, make B the subject of this formula.

2

- (b) The length of each edge of a cube is x cm.

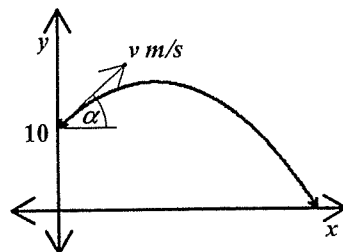
- (i) Write expressions for the surface area (A) and volume (V) of the cube and hence find $\frac{dA}{dx}$ and $\frac{dV}{dx}$.

1

- (ii) The surface area of the cube increases at a rate of $6 \text{ cm}^2/\text{second}$. Find the rate of change of the volume of the cube when the length of each edge is 5 cm.

2

- (c) A stone is thrown from a 10m high cliff with velocity $v \text{ m/s}$ at an angle of projection α . The stone's horizontal displacement from the origin, t seconds after being thrown, is given by the equation $x = vt \cos \alpha$. Do not prove this.



- (i) Given that $\ddot{y} = -g$, prove that the stone's vertical displacement from the origin, t seconds after being thrown, is given by $y = vt \sin \alpha - \frac{gt^2}{2} + 10$.

3

- (ii) Show that $y = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2v^2} + 10$.

1

- (iii) Given that $\alpha = 45^\circ$, $g = 10 \text{ m/s}^2$ and $v = 15 \text{ m/s}$, how far from the base of the cliff does the stone land?

3

Question 7 (12 marks) [BEGIN A NEW PAGE]

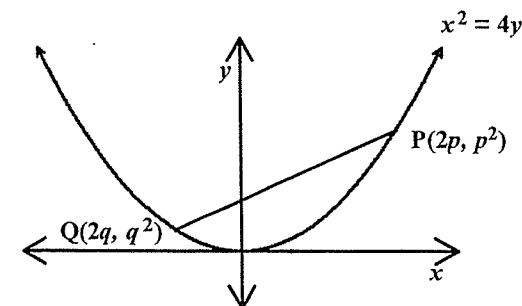
Marks

- (a) It is known that if $\frac{dy}{dx} = y$, then $y = e^x$ is a solution (since $\frac{dy}{dx} = e^x = y$).

If $\frac{dy}{dx} = \frac{1}{y}$, find a solution for y .

2

- (b)



- (i) Show that the gradient of PQ is $\frac{p+q}{2}$.

1

For the remainder of the question, assume that PQ is a focal chord, passing through the focus $F(0, 1)$.

- (ii) Show that $pq = -1$.

2

- (iii) Show that the equation of PQ is $y = \left(\frac{p^2 - 1}{2p} \right)x + 1$.

1

- (iv) Let A be the area bounded by the parabola and the focal chord.

Show that $A = \frac{1}{3} \left(p^3 + 3p + \frac{3}{p} + \frac{1}{p^3} \right)$.

[You may assume that $p > 0$ and $q < 0$]

3

- (v) Hence, or otherwise, find the value of p that gives the minimum area found in part (iv).

3

End of Paper

① (a)

$$A(-2, 4) \quad B(10, -8)$$

$$3 \quad -2$$

$$P = \left(\frac{3 \times 10 - 2 \times (-2)}{3 - 2}, \frac{3 \times (-8) - 2 \times 4}{3 - 2} \right)$$

$$= (34, -32)$$

1 mark for
internal division
with correct
answer $(5\frac{1}{2}, -3\frac{1}{2})$

(b)

$$\int x \sqrt{2x-1} \, dx \quad u = 2x-1$$

$$= \int \frac{(u+1)}{2} \cdot u^{\frac{1}{2}} \cdot \frac{du}{2}$$

$$= \frac{1}{4} \int (u^{\frac{3}{2}} + u^{\frac{1}{2}}) du$$

$$= \frac{1}{4} \left[\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right] + C$$

$$= \frac{1}{10} (2x-1)^{\frac{5}{2}} + \frac{1}{6} (2x-1)^{\frac{3}{2}} + C$$

No penalty for
not having a
constant of integration

(c) The statement is false as there
are far more day students
than boarders which means you
are more likely to randomly
select a day student.

Must justify
answer to get
the mark.

(d) (i) $m_{PA} = m_{PB} - 1$

$$\therefore \frac{y}{x+3} = \frac{y}{x-3} - 1$$

$$y(x-3) = y(x+3) - (x+3)(x-3)$$

$$xy - 3y = xy + 3y - (x^2 - 9)$$

$$x^2 - 9 = 6y$$

$$x^2 = 6y + 9$$

$$(ii) \quad x^2 = 4(1.5)(y+1.5)$$

$\therefore$ vertex is $(0, -1.5)$ & focal length = $1.5u$

$\therefore$ Focus is the origin

(1)

$$(e) \quad \frac{1}{x^3} > \frac{1}{x^5}, x \neq 0$$

$$x^6 \times \frac{1}{x^3} > x^6 \times \frac{1}{x^5} \quad \checkmark \leftarrow \text{can also multiply by } x^8, x^{10}, \text{ etc.}$$

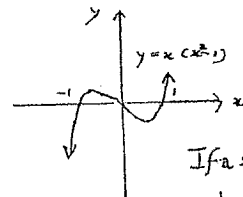
$$x^3 > x$$

$$x^3 - x > 0$$

$$x(x^2 - 1) > 0$$

$$x(x+1)(x-1) > 0 \quad \checkmark$$

$$\therefore -1 < x < 0 \text{ or } x > 1 \quad \checkmark$$



Ignore any irregularities
with $\leq$ or $\geq$ instead of
< and >.

If a student multiplies thru' by x^5
and gets $x < -1$ or $x > 1$ award
a total of 1 mark.

(2)

$$(a) \quad \sin(-15^\circ)$$

$$= \sin(30^\circ - 45^\circ)$$

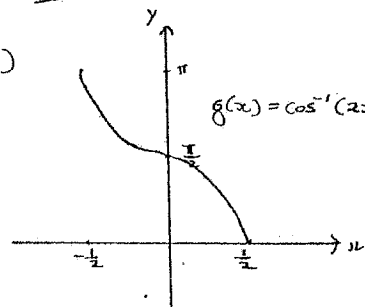
$$= \sin 30^\circ \cos 45^\circ - \cos 30^\circ \sin 45^\circ \quad \checkmark$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$\checkmark$ - ignore subsequent errors
 $\downarrow$

(b) (i)



correct domain 1 mark
correct shape & range 1 mark

$$(ii) \quad \text{Let } f(x) = g(x)$$

$$\text{i.e. } \cos^{-1}(2x) = \sin^{-1}(x)$$

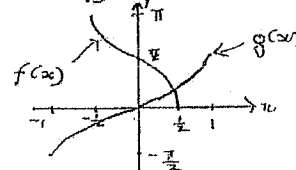
$$\text{Now } \sin^2 \alpha + \cos^2 \alpha = 1$$

$$\therefore x^2 + (2x)^2 = 1 \quad \checkmark$$

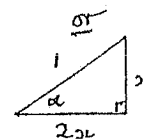
$$5x^2 = 1$$

$$x^2 = \frac{1}{5}$$

$$x = \frac{1}{\sqrt{5}}, \text{ since } x > 0 - \text{see graph} \leftarrow \text{ignore } \pm \text{ in marking.}$$



$$\left. \begin{array}{l} \text{Let } \sin^{-1} x = \alpha \\ \Rightarrow x = \sin \alpha \\ \text{Let } \cos^{-1}(2x) = \alpha \\ \Rightarrow 2x = \cos \alpha \end{array} \right\} \quad \checkmark$$



$$x^2 + (2x)^2 = 1 \quad \checkmark$$

etc.

②
(c)

$$(ii) f'(x) = \frac{-2}{\sqrt{1-(2x)^2}} \quad \checkmark$$

$$f'(\frac{1}{\sqrt{5}}) = \frac{-2}{\sqrt{1-\frac{4}{5}}} \\ = \frac{-2}{\frac{1}{\sqrt{5}}} \quad \checkmark \\ = -2\sqrt{5}$$

$$g'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$g'(\frac{1}{\sqrt{5}}) = \frac{1}{\sqrt{1-\frac{1}{5}}} \\ = \frac{1}{\sqrt{\frac{4}{5}}} \\ = \frac{1}{\frac{2}{\sqrt{5}}} \quad \checkmark \\ = \frac{\sqrt{5}}{2}$$

$$(iv) \tan(\beta - \alpha) = \frac{\tan \beta - \tan \alpha}{1 + \tan \beta \tan \alpha} \quad \checkmark$$

(v) Let the required angle be ϕ

$$\tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{\sqrt{5}}{2} + 2\sqrt{5}}{1 - 5} \right| \quad \checkmark$$

$$= \frac{5\sqrt{5}}{8}$$

$$\therefore \phi = 54.41\dots^\circ$$

$$= \underline{\underline{54^\circ}} \text{ (n.deg.)} \quad \checkmark \text{ Don't worry about accuracy.}$$

③ (a) (i) $y = \ln(\sin x)$

$$\frac{dy}{dx} = \frac{\cos x}{\sin x} \quad \checkmark$$

$$= \cot x$$

$$(ii) \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \cot x \, dx$$

$$= \left[\ln(\sin x) \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \quad \checkmark$$

$$= \ln(\sin \frac{3\pi}{4}) - \ln(\sin \frac{\pi}{4})$$

$$= \ln(\frac{1}{\sqrt{2}}) - \ln(\frac{1}{\sqrt{2}})$$

$$= \underline{\underline{0}} \quad \checkmark$$

$$(b) \int_0^{\frac{\pi}{12}} \cos^2(3x) \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{12}} (\cos 6x + 1) \, dx \quad \checkmark$$

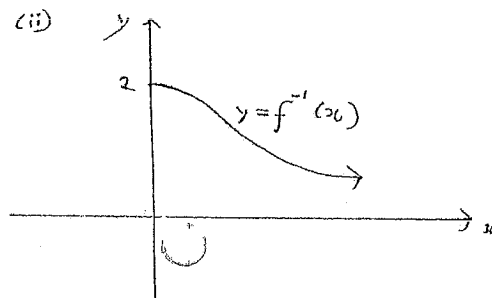
$$= \frac{1}{2} \left[\frac{\sin 6x}{6} + x \right]_0^{\frac{\pi}{12}} \quad \checkmark$$

$$= \frac{1}{2} \left\{ \left(\frac{1}{6} + \frac{\pi}{12} \right) - (0+0) \right\} \quad \checkmark$$

$$= \underline{\underline{\frac{2+\pi}{24}}}$$

$$(c) (i) \left. \begin{array}{l} D_{f^{-1}} : x \geq 0 \\ R_{f^{-1}} : 0 < y \leq 2 \end{array} \right\} 1$$

Must have inequality signs exactly right for the mark.



y-intercept and shape required for 1 mark.

③(c)
(iii)

$$y = f(x) = \sqrt{\frac{2-x}{x}}$$

$$\therefore x = \sqrt{\frac{2-y}{y}} \text{ is inverse function } \checkmark$$

$$x^2 = \frac{2-y}{y}$$

$$\text{or } x^2 y = 2-y$$

$$x^2 = \frac{2}{y} - 1 \checkmark$$

$$x^2 y + y = 2$$

$$x^2 + 1 = \frac{2}{y}$$

$$y(x^2 + 1) = 2 \checkmark$$

$$y = \frac{2}{x^2 + 1} = f^{-1}(x)$$

(iv)

$$\text{Consider } \int_0^a \frac{2}{1+x^2} dx = 2 [\tan^{-1} x]_0^a$$

$$= 2 \{ \tan^{-1} a - \tan^{-1} 0 \}$$

$$= 2 \tan^{-1} a$$

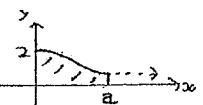
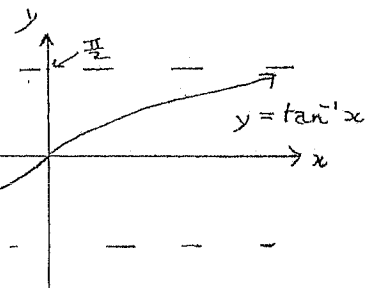
$$\text{Req'd area} = \lim_{a \rightarrow \infty} 2 \tan^{-1} a$$

$$= 2 \lim_{a \rightarrow \infty} \tan^{-1} a$$

$$= 2 \times \frac{\pi}{2} \checkmark$$

$$= \pi \text{ units}^2$$

Noting:



$$\text{or } A = \int_0^{\infty} \frac{2}{1+x^2} dx \checkmark$$

$$= 2 [\tan^{-1} x]_0^{\infty}$$

$$= 2 \{ \tan^{-1} \infty - \tan^{-1} 0 \}$$

$$= 2 \{ \frac{\pi}{2} - 0 \} \checkmark$$

$$= \pi \text{ units}^2$$

④(a)

$$(i) y = \frac{(x-\sqrt{3})(x+\sqrt{3})}{x+2}$$

$$\Rightarrow x = -2 \text{ is vertical asymptote}$$

$$\text{when } x = 0, y = -\frac{3}{2} \text{ [y-intercept]}$$

$$\text{when } y = 0, x = \pm \sqrt{3} \text{ [x-intercepts]} \checkmark$$

$$(ii) \frac{dy}{dx} = \frac{(x+2)2x - (x^2-3)1}{(x+2)^2} \checkmark$$

$$= \frac{2x^2 + 4x - x^2 + 3}{(x+2)^2}$$

$$= \frac{x^2 + 4x + 3}{(x+2)^2}$$

$$= \frac{(x+1)(x+3)}{(x+2)^2}$$

$$\Rightarrow \text{stat. pts at } x = -1, -3 \checkmark$$

$$\text{when } x = -1, y = \frac{-1-3}{-1} = -2$$

$$\text{when } x = -3, y = \frac{-3-3}{-1} = -6$$

test (-1, -2):

x	$-1\frac{1}{2}$	-1	0
$\frac{dy}{dx}$	$\frac{(-\frac{1}{2})(\frac{1}{2})}{(\frac{1}{2})^2}$	0	$\frac{3}{4}$
	neg.		pos.

min. t.p.

test (-3, -6):

x	-4	-3	$-2\frac{1}{2}$
$\frac{dy}{dx}$	$\frac{(-3)(-1)}{(-2)^2}$	0	$\frac{(-1\frac{1}{2})(\frac{1}{2})}{(-\frac{1}{2})^2}$
	pos.		neg.

max. t.p.

$$(iii) (x-2) + \frac{1}{x+2} = \frac{(x-2)(x+2) + 1}{x+2}$$

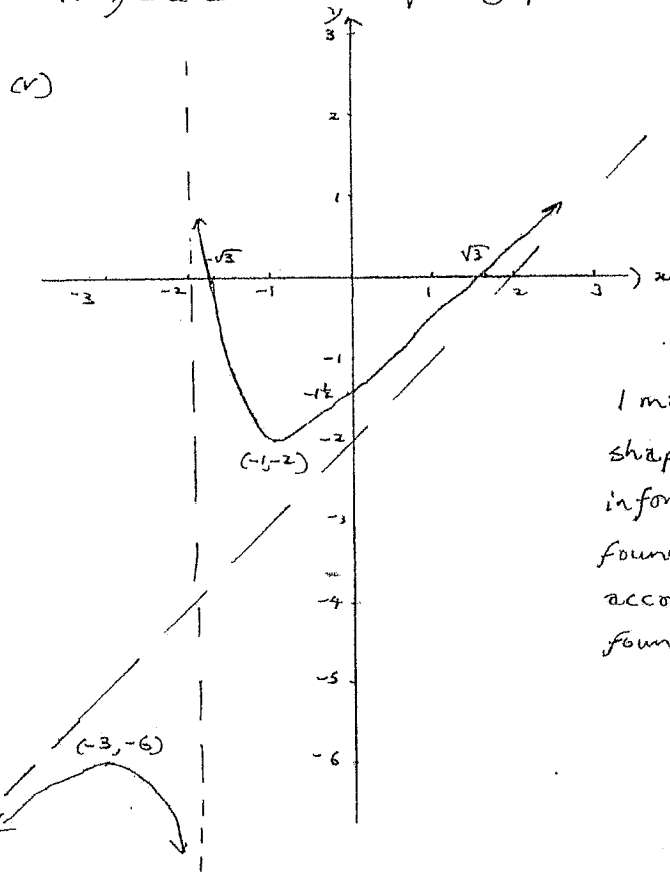
$$= \frac{x^2 - 4 + 1}{x+2} \checkmark$$

$$= \frac{x^2 - 3}{x+2}$$

④ (a)(iv)

$$\text{as } x \rightarrow \infty, (x-2) + \frac{1}{x+2} \rightarrow x-2$$

$\therefore y = x-2$ is an oblique asymptote. ✓



1 mark for correct shape - (all other information has been found earlier) - according to values found above.

b) when $n=0$, $5^0+3=4 \nmid 2$

$$\Rightarrow 5^n+3 \nmid 2 \text{ for } n=0. \quad \checkmark$$

Assume $5^k+3=2J, J \in \mathbb{Z}^+$

i.e. $5^k=2J-3$

Required to prove $5^{k+1}+3 \nmid 2$

Now $5^{k+1}+3 = 5 \cdot 5^k+3$ ✓
 $= 5(2J-3)+3$ by induction
 $= 10J-12$ supposition

④ (b) (cont.)

$$\Rightarrow 5^{k+1}+3 \nmid 2 \text{ if } 5^k+3 \nmid 2 \quad (*)$$

Since $5^n+3 \nmid 2$ for $n=0$
it is true for $n=1$ and
hence true for all subsequent
positive integral values of n
by (*).

⑤ (a)(i) $x = 5 \cos(2t) - 12 \sin(2t)$

$$\dot{x} = -10 \sin(2t) - 24 \cos(2t) \quad \checkmark$$

$$\begin{aligned} \ddot{x} &= -20 \cos(2t) + 48 \sin(2t) \\ &= -4(5 \cos(2t) - 12 \sin(2t)) \\ &= -4x \quad \checkmark \end{aligned}$$

which is in the form $\ddot{x} = -n^2 x$.
Hence the motion is simple harmonic.

(ii) Period $= \frac{2\pi}{2} = \pi$ seconds ✓

(iii) $R \cos(2t + \alpha)$
 $= R \{ \cos(2t) \cos \alpha - \sin(2t) \sin \alpha \}$

$$\therefore \text{Let } 5 \cos(2t) - 12 \sin(2t) \equiv R \cos(2t) \cos \alpha - R \sin(2t) \sin \alpha$$

$$\Rightarrow R \cos \alpha = 5 \quad \& \quad R \sin \alpha = 12 \quad \checkmark$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 5^2 + 12^2$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 169$$

$$R = 13 \quad [R > 0]$$

$$\frac{R \sin \alpha}{R \cos \alpha} = \frac{12}{5} \Rightarrow \tan \alpha = \frac{12}{5} \quad \checkmark$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{12}{5}\right) \quad [\alpha \text{ acute}]$$

$$\therefore x = 13 \cos\left(2t + \tan^{-1}\left(\frac{12}{5}\right)\right)$$

also $\alpha = \sin^{-1}\left(\frac{12}{13}\right)$
 $= \cos^{-1}\left(\frac{5}{13}\right)$

1 mark for getting both R and α .

3(a)

(iv) Amplitude = 13 metres. ✓

(v) Particle is at centre of motion

when acceleration is zero

at $x = 0$ as eqn is in

form $\ddot{x} = -n^2x$

$$\text{i.e. when } 5 \cos(2t) - 12 \sin(2t) = 0$$

$$\text{i.e. when } 13 \cos(2t + \tan^{-1}(\frac{12}{5})) = 0 \quad \checkmark$$

$$\Rightarrow \cos(2t + \tan^{-1}(\frac{12}{5})) = 0$$

$$2t + \tan^{-1}(\frac{12}{5}) = 2n\pi \pm \cos^{-1}(0)$$

$$2t + \tan^{-1}(\frac{12}{5}) = 2n\pi \pm \frac{\pi}{2} \quad \checkmark$$

$$t = \frac{2n\pi + \frac{\pi}{2} - \tan^{-1}(\frac{12}{5})}{2}, \frac{2n\pi - \frac{\pi}{2} - \tan^{-1}(\frac{12}{5})}{2}$$

choosing n to be an integer such that $t > 0$.

(b)(i) $\angle ADB = 75^\circ$ ($\angle$ sum Δ)

$$\therefore \frac{AB}{h} = \tan 75^\circ \quad \text{or} \quad \frac{h}{AB} = \tan 15^\circ \quad \text{or} \quad AB = \frac{h}{\tan 15^\circ}$$

$$\therefore AB = h \tan 75^\circ \quad \checkmark \quad AB = h \cot 15^\circ$$

(ii) Similarly $BC = h \tan 78^\circ$

By Pythagoras: $AB^2 + AC^2 = BC^2$

$$\therefore h^2 \tan^2 75^\circ + 500^2 = h^2 \tan^2 78^\circ \quad \checkmark$$

$$h^2 (\tan^2 78^\circ - \tan^2 75^\circ) = 250000$$

$$h = \frac{500}{\sqrt{\tan^2 78^\circ - \tan^2 75^\circ}} \quad \checkmark$$

$$= 174.55 \dots$$

$$= \underline{175 \text{ m (nearest metre)}} \quad \checkmark \quad \text{No penalty for accuracy/units.}$$

$$6(a) \quad t = -\frac{1}{A} \ln\left(\frac{B-C}{D}\right)$$

$$-At = \ln\left(\frac{B-C}{D}\right)$$

$$e^{-At} = \frac{B-C}{D} \quad \checkmark$$

$$B-C = D e^{-At}$$

$$\therefore B = C + D e^{-At} \quad \checkmark$$

$$(b)(i) \quad A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{1 mark}$$

$$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$$

← If any of this done in (ii) give mark.

$$(ii) \quad \frac{dA}{dt} = 6 \text{ cm}^2/\text{s}$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$\text{or} \quad \frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt} \times \frac{dA}{dt}$$

$$= 3x^2 \times \frac{1}{12x} \times 6 \quad \checkmark$$

$$\text{when } x = 5, \quad 6 = (12 \times 5) \times \frac{dx}{dt}$$

$$\text{when } x = 5, \quad \frac{dV}{dt} = 3 \times 5^2 \times \frac{1}{12 \times 5} \times 6 \quad \checkmark$$

$$\frac{dx}{dt} = \frac{1}{10} \text{ cm/s} \quad \checkmark$$

$$= 7.5 \text{ cm}^3/\text{s}$$

$$\text{and} \quad \frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$$

$$= (3 \times 5^2) \times \frac{1}{10} \quad \checkmark$$

$$= \underline{7.5 \text{ cm}^3/\text{s}}$$

$$\text{or} \quad \frac{dx}{dt} = \frac{dx}{dA} \times \frac{dA}{dt}$$

$$= \frac{1}{12x} \times 6$$

$$= \frac{1}{2x} \quad \checkmark$$

$$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$$

$$= 3x^2 \times \frac{1}{2x}$$

$$= \frac{3x}{2}$$

$$\text{when } x = 5, \quad \frac{dV}{dt} = \frac{3 \times 5}{2} \quad \checkmark$$

$$= 7.5 \text{ cm}^3/\text{s}$$

(c)

$$\ddot{y} = -g$$

$$\dot{y} = \int -g dt$$

$$= -gt + c_1$$

$$\text{when } t = 0, \quad \dot{y} = v \sin \alpha$$

$$v \sin \alpha = c_1 \quad \checkmark$$

$$\therefore \dot{y} = -gt + v \sin \alpha$$

$$y = \int (-gt + v \sin \alpha) dt$$

$$= -\frac{gt^2}{2} + vt \sin \alpha + c_2 \quad \checkmark$$

$$\text{when } t = 0, \quad y = 10$$

$$10 = c_2 \quad \checkmark$$

$$\therefore y = vt \sin \alpha - \frac{gt^2}{2} + 10$$

Students must demonstrate by substitution how they arrive at both the constants of integration - do not award full marks to solutions that skip any key steps in the process.

⑥(c)(ii)

$$x = vt \cos \alpha \Rightarrow t = \frac{x}{v \cos \alpha}$$

$$\therefore y = v \times \frac{x}{v \cos \alpha} \times \sin \alpha - \frac{g \times \left(\frac{x}{v \cos \alpha}\right)^2}{2} + 10 \checkmark$$

$$= x \tan \alpha - \frac{g x^2}{2 v^2 \cos^2 \alpha} + 10$$

$$= x \tan \alpha - \frac{g x^2 \sec^2 \alpha}{2 v^2} + 10$$

(iii) $\alpha = 45^\circ, g = 10 \text{ m/s}^2, v = 15 \text{ m/s}$
 $\Rightarrow y = x - \frac{10 x^2 \times 2}{2 \times 225} + 10 \checkmark$ [substituting & calculating the value of $\sec^2 45^\circ$]

When $y = 0, 0 = x - \frac{10 x^2}{225} + 10$

$$\frac{10 x^2}{225} - x + 10 = 0$$

$$\frac{2 x^2}{45} - x + 10 = 0$$

$$2 x^2 - 45 x + 450 = 0 \checkmark$$

$$(2x + 15)(x - 30) = 0$$

$\therefore x = 30 (x > 0) \checkmark$ ← Must exclude other

So stone lands 30m from base of cliff. solution to quadratic.

⑦ (a) $\frac{dx}{dy} = y$ if $\frac{dy}{dx} = \frac{1}{y}$

$$\therefore \int \frac{dx}{dy} dy = \int y dy \checkmark$$

$$x = \frac{y^2}{2} + \text{a constant}$$

$$y^2 = 2x + \text{a constant}$$

$\therefore y = \pm \sqrt{2x} + c \checkmark \rightarrow$ accept $y = \sqrt{2x}$
 or $y = -\sqrt{2x}$
 {with or without a constant}

(b) (i) $m_{PQ} = \frac{p^2 - q^2}{2p - 2q}$
 $= \frac{(p+q)(p-q)}{2(p-q)} \checkmark$
 $= \frac{p+q}{2}$

⑦(b)

(ii) Since PQ passes through F,

$$m_{PF} = m_{PQ}$$

$$\therefore \frac{p^2 - 1}{2p} = \frac{p+q}{2} \checkmark$$

$$\cancel{2p^2 - 2} = \cancel{2p^2} + 2pq \checkmark$$

$$\therefore pq = -1$$

(iii) As above $m_{PQ} = m_{PF}$

$$= \frac{p^2 - 1}{2p}$$

OR $y - 1 = \left(\frac{p+q}{2}\right)(x - 0)$

$$y = \left(\frac{p+q}{2}\right)x + 1$$

$$= \left(\frac{p - \frac{1}{p}}{2}\right)x + 1 \checkmark$$

$$= \left(\frac{p^2 - 1}{2p}\right)x + 1$$

Also PQ passes through (0, 1)

so its y-intercept is 1. ← both for mark

Hence using $y = mx + b$,

$$PQ \text{ is } y = \left(\frac{p^2 - 1}{2p}\right)x + 1.$$

(iv) $x^2 = 4y \Rightarrow y = \frac{x^2}{4}$

$$\& pq = -1 \Rightarrow q = -\frac{1}{p}$$

$$\therefore A = \int_{-\frac{2}{p}}^{2p} \left\{ \left(\frac{p^2 - 1}{2p}\right)x + 1 - \frac{x^2}{4} \right\} dx \checkmark$$

$$= \left[\left(\frac{p^2 - 1}{4p}\right)x^2 + x - \frac{x^3}{12} \right]_{-\frac{2}{p}}^{2p}$$

$$= \left\{ \left(\frac{p^2 - 1}{4p}\right)4p^2 + 2p - \frac{8p^3}{12} \right\} - \left\{ \left(\frac{p^2 - 1}{4p}\right)\frac{4}{p^2} - \frac{2}{p} + \frac{8}{12p^3} \right\} \checkmark$$

$$= p^3 - p + 2p - \frac{2p^3}{3} - \frac{1}{p} + \frac{1}{p^3} + \frac{2}{p} - \frac{2}{3p^3}$$

$$= \frac{p^3}{3} + p + \frac{1}{p} + \frac{1}{3p^3} \checkmark$$

$$= \frac{1}{3} \left(p^3 + 3p + \frac{3}{p} + \frac{1}{p^3} \right)$$

⑦ (b)
(iv) (cont.)

Alternatively:

$$\begin{aligned}
 A &= \int_{2q}^{2p} \left\{ \left(\frac{p^2-1}{2p} \right) x + 1 - \frac{x^2}{4} \right\} dx \\
 &= \left[\left(\frac{p^2-1}{4p} \right) x^2 + x - \frac{x^3}{12} \right]_{2q}^{2p} \quad \checkmark \\
 &= \left\{ \left(\frac{p^2-1}{4p} \right) \times 4p^2 + 2p - \frac{8p^3}{12} \right\} - \left\{ \left(\frac{p^2-1}{4p} \right) \times 4q^2 + 2q - \frac{8q^3}{12} \right\} \\
 &= p^3 - p + 2p - \frac{2p^3}{3} - pq^2 + \frac{q^2}{p} - 2q + \frac{2q^3}{3} \\
 &= \frac{p^3}{3} + p - p \times \left(-\frac{1}{p} \right)^2 + \frac{\left(-\frac{1}{p} \right)^2}{p} - 2 \left(-\frac{1}{p} \right) + \frac{2 \left(-\frac{1}{p} \right)^3}{3} \quad \left\{ \begin{array}{l} \text{since} \\ q = -\frac{1}{p} \end{array} \right\} \quad \checkmark \\
 &= \frac{p^3}{3} + p - \frac{1}{p} + \frac{1}{p^3} + \frac{2}{p} - \frac{2}{3p^3} \\
 &= \frac{p^3}{3} + p + \frac{1}{p} + \frac{1}{3p^3} \quad \checkmark \\
 &= \frac{1}{3} \left(p^3 + 3p + \frac{3}{p} + \frac{1}{p^3} \right)
 \end{aligned}$$

$$(v) \quad A = \frac{1}{3} (p^3 + 3p + 3p^{-1} + p^{-3})$$

$$\begin{aligned}
 \frac{dA}{dp} &= \frac{1}{3} (3p^2 + 3 - 3p^{-2} - 3p^{-4}) \quad \checkmark \\
 &= p^2 + 1 - \frac{1}{p^2} - \frac{1}{p^4}
 \end{aligned}$$

$$\text{when } \frac{dA}{dp} = 0, \quad p^2 + 1 - \frac{1}{p^2} - \frac{1}{p^4} = 0$$

$$\therefore p^6 + p^4 - p^2 - 1 = 0$$

$$p^4(p^2+1) - 1(p^2+1) = 0$$

$$(p^4-1)(p^2+1) = 0 \quad \checkmark$$

$$\therefore p^4-1=0 \quad \left\{ \begin{array}{l} \text{no real solns} \\ \text{for } p^2+1=0 \end{array} \right\}$$

$$\therefore p^4=1 \Rightarrow p=1 \quad (p>0)$$

$$\begin{aligned}
 \text{when } p = \frac{1}{2}, \quad \frac{dA}{dp} &= \left(\frac{1}{2} \right)^2 + 1 - \frac{1}{\left(\frac{1}{2} \right)^2} - \frac{1}{\left(\frac{1}{2} \right)^4} \\
 &= \frac{1}{4} + 1 - 4 - 16 < 0
 \end{aligned}$$

$$\text{when } p=2, \quad \frac{dA}{dp} = 4 + 1 - \frac{1}{4} - \frac{1}{16} > 0$$

min. at p